

Arithmophysics: Source Truncation and Growing Mode Bounds

Analytical continuation of the shared checkpoint

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Strongest new estimate in this continuation. Fix absolute constants $A > 0$, $0 < b \leq 1$ for the established unconditional energy bound

$$R(L) \leq ALe^{-b\sqrt{\log(eL)}} \quad (L \geq 1).$$

Put $c = b/2$ and $C = (\sqrt{2A} + 3/2)^2$. Then for every integer $N \geq 1$ and every $1 \leq J \leq N$,

$$\max \left\{ \lambda_{J,N} S_N(J), \sum_{j=J}^N \lambda_{j,N} |b_N(j)|^2 \right\} \leq C \frac{N}{J} e^{-c\sqrt{\log(eN/J)}}.$$

The earlier bound by $R(N)$ remains valid simultaneously. Consequently,

$$S_N(J) \leq \pi^2 \min \left\{ A \frac{J^2}{N} e^{-b\sqrt{\log(eN)}}, C \frac{J}{N} e^{-c\sqrt{\log(eN/J)}} \right\}.$$

This strengthens the inherited mode estimate on growing ranges. For example, at $J \asymp \sqrt{N}$, the new population bound has the factor $N^{-1/2}$, in addition to an exponential saving in $\sqrt{\log N}$.

Mechanism and arithmetic input. Truncate the actual completed Möbius source at L . Its energy remains exactly $R(L)$ in the larger horizon. The remaining squared coefficient norm is at most $1/L$. The signed interaction between these two pieces is retained and bounded by Cauchy–Schwarz in the stated projected norm. Optimizing L produces the estimate above. The arithmetic input is the existing unconditional bound on $R(L)$; no new growth hypothesis is assumed.

Closing status. For any prescribed tolerance N^η , $0 < \eta < 1$, all mode cutoffs $J \geq \lceil N^{1-\eta} \rceil$ now satisfy that tolerance, and so does the entire energy in modes above that index. The remaining instance of (PE) is restricted to smaller cutoffs, as specified in Section 6. The all-scale subpower bound for the complete $R(N)$ is not proved here.

The growth law (G) remains an explicit premise in the larger exploratory framework. This closing attempt uses independent established inputs.

1 The candidate and the exact source being estimated

Let $c_N(n) = \mu(n)/n$, $b_N = V_N^* c_N$, and

$$v_{j,N}(n) = \frac{2}{\sqrt{2N+1}} \sin \frac{n(2j-1)\pi}{2N+1}, \quad \lambda_{j,N} = \frac{1}{4 \sin^2((2j-1)\pi/(4N+2))}.$$

The matrix $B_N(a, b) = \min(a, b)$ satisfies $B_N = V_N \Lambda_N V_N^*$. The full energy, including its terminal, is

$$R(N) = c_N^* B_N c_N = \sum_{k \leq N} \frac{M(k)^2}{k(k+1)} + \frac{M(N)^2}{N}.$$

We estimate the unweighted source population $S_N(J) = \sum_{j \leq J} |b_N(j)|^2$ and the energy tail

$$E_N^{\geq J} = \sum_{j=J}^N \lambda_{j,N} |b_N(j)|^2.$$

These concern the source $\mu(n)/n$. The log-weighted vector F_N and the continuous coordinates z_j from the shared checkpoint retain their own definitions and are not identified with b_N .

Candidate upper comparison. The estimate attempted is

$$\lambda_{J,N} S_N(J) \leq C \frac{N}{J} e^{-c\sqrt{\log(eN/J)}}.$$

Its proposed independent input is the already established all-horizon estimate

$$R(L) \leq A L e^{-b\sqrt{\log(eL)}} \quad (L \geq 1). \quad (1)$$

Such absolute $A, b > 0$ exist by the classical unconditional Möbius bound $M(x) = O(xe^{-a\sqrt{\log x}})$ and the historical-plus-terminal identity. Splitting that identity at $\sqrt{L}$ proves the claimed form after reducing the exponential constant and increasing A to cover small L . We may take $b \leq 1$ and $A \geq e^b$. No numerical fit of these constants is used.

For the source diagonal and off-diagonal, keep the complete weighted pair sum:

$$K_{N,J}(a, b) = \sum_{j \leq J} v_{j,N}(a) v_{j,N}(b), \quad \mathcal{O}_{N,J} = 2 \sum_{a < b \leq N} \frac{\mu(a)\mu(b)}{ab} K_{N,J}(a, b),$$

$$\mathcal{D}_{N,J} = \sum_{n \leq N} \frac{\mu(n)^2}{n^2} K_{N,J}(n, n).$$

The established diagonal allowance gives

$$S_N(J) = \mathcal{D}_{N,J} + \mathcal{O}_{N,J}, \quad \mathcal{D}_{N,J} \leq \frac{\pi^2}{2} \frac{J^2}{N^2}, \quad (\mathcal{O}_{N,J})_+ \leq S_N(J).$$

Thus the candidate, if established, gives an independent upper bound for the positive part of the complete signed off-diagonal. No universal negative sign is used. The source cutoff L below is chosen from N, J ; the earlier fixed inverse-localization cutoff 21 remains a separate established tool.

2 Exact truncation inequality with every mixed term

For $0 \leq L \leq N$, let $c^{\leq L}$ be c_N truncated to its first L entries, and $c^{>L} = c_N - c^{\leq L}$. Put

$$h = V_N^* c^{\leq L}, \quad t = V_N^* c^{>L}, \quad b_N = h + t, \quad T_{L,N} = \sum_{L < n \leq N} \frac{\mu(n)^2}{n^2}.$$

The symbol $T_{L,N}$ here denotes a scalar coefficient norm; integer-dilation matrices keep their earlier subscripted notation in the preceding papers. Set $R(0) = 0$. For $L \geq 1$, $T_{L,N} \leq 1/L$, while $T_{N,N} = 0$.

Theorem 2.1. *For every $N \geq 1$, $1 \leq J \leq N$, and $0 \leq L \leq N$,*

$$\boxed{\max\{\lambda_{J,N} S_N(J), E_N^{\geq J}\} \leq \left[\sqrt{R(L)} + \sqrt{\lambda_{J,N} T_{L,N}} \right]^2.} \quad (2)$$

For a different coefficient sequence a_n/n , the same inequality holds with its own prefix energy and its own squared coefficient tail.

Proof. The leading $L \times L$ submatrix of B_N is exactly B_L . Therefore

$$\sum_j \lambda_{j,N} |h_j|^2 = R(L), \quad \sum_j |t_j|^2 = T_{L,N}.$$

For the first J coordinates, decreasing gains give $\lambda_{J,N} \|\Pi_J h\|_2^2 \leq R(L)$ and $\lambda_{J,N} \|\Pi_J t\|_2^2 \leq \lambda_{J,N} T_{L,N}$. The exact signed identity is

$$S_N(J) = \|\Pi_J h\|^2 + \|\Pi_J t\|^2 + 2 \operatorname{Re} \langle \Pi_J h, \Pi_J t \rangle.$$

Cauchy–Schwarz on this mixed term proves the first bound. For $j \geq J$, the prefix’s weighted energy is at most $R(L)$, and the tail’s weighted energy is at most $\lambda_{J,N} T_{L,N}$. The same argument in the weighted norm proves the second bound. $\square$

All internal interactions within the prefix remain in $R(L)$, including $M(L)^2/L$. The remaining source is the actual coefficient tail. Its *unweighted coefficient norm* is bounded by $1/L$; the gain $\lambda_{J,N}$ is essential when it is used in the energy comparison.

This is a direct estimate on the original source Gram form. It does not replace the Möbius coefficients by independent signs. The arithmetic improvement enters only when the actual prefix energy is bounded using (1).

3 Optimizing the cutoff at every horizon

Theorem 3.1. *Assume the established bound (1). Define*

$$c = b/2, \quad C = (\sqrt{2A} + 3/2)^2, \quad \tau = N/J.$$

For every $N \geq 1$ and $1 \leq J \leq N$,

$$\max\{\lambda_{J,N} S_N(J), E_N^{\geq J}\} \leq \min\{ANe^{-b\sqrt{\log(eN)}}, C\tau e^{-c\sqrt{\log(e\tau)}}\}. \quad (3)$$

Proof. The first term follows because both quantities are at most $R(N)$. For the second put $v = \log(e\tau)$ and $L_* = \tau e^{c\sqrt{v}}$.

If $L_* \leq N$, choose the integer $L = \lceil L_* \rceil$. Then $L_* \leq L \leq 2L_*$, and (1) gives

$$R(L) \leq 2A\tau e^{(c-b)\sqrt{v}} = 2A\tau e^{-c\sqrt{v}}.$$

The uniform gain bound $\lambda_{J,N} \leq 9N^2/(4J^2)$ and the coefficient tail estimate give

$$\lambda_{J,N} T_{L,N} \leq \frac{9}{4} \frac{\tau^2}{L_*} = \frac{9}{4} \tau e^{-c\sqrt{v}}.$$

Apply Theorem 2.1. The resulting squared constant is precisely C .

If $L_* > N$, choose $L = N$, so the coefficient tail vanishes exactly. The cutoff inequality is $J < e^{c\sqrt{v}}$. Since $v \leq \log(eN)$,

$$\begin{aligned} \frac{R(N)}{\tau e^{-c\sqrt{v}}} &\leq AJ \exp[-2c\sqrt{\log(eN)} + c\sqrt{v}] \\ &\leq A \exp[2c\sqrt{v} - 2c\sqrt{\log(eN)}] \leq A \leq C. \end{aligned}$$

This proves the second bound even when the nominal balancing cutoff would exceed the horizon. Rounding and all small horizons are included. $\square$

Using $\lambda_{J,N} \geq N^2/(\pi^2 J^2)$ yields

$$S_N(J) \leq \pi^2 \min \left\{ A \frac{J^2}{N} e^{-b\sqrt{\log(eN)}}, C \frac{J}{N} e^{-c\sqrt{\log(eN/J)}} \right\}. \quad (4)$$

The complete source off-diagonal consequently satisfies $(\mathcal{O}_{N,J})_+ \leq S_N(J)$ with this same upper bound. The result is uniform, rather than an average over horizons or an estimate outside an exceptional set.

4 The quantitative gain and the role of arithmetic

For fixed $0 < \alpha < 1$ and $J \asymp N^\alpha$, the additional estimate is

$$S_N(J) \ll N^{\alpha-1} e^{-c\sqrt{(1-\alpha)\log N}}, \quad E_N^{\geq J} \ll N^{1-\alpha} e^{-c\sqrt{(1-\alpha)\log N}}.$$

The ratio of the new population envelope to the inherited (BL) envelope is $N^{-\alpha+o(1)}$. Both estimates are preserved in (4); the new one never displaces a stronger instance of (BL). When (BL) exceeds the independent Parseval bound, the new estimate also improves on that bound because $N^{\alpha-1}$ tends to zero.

Mode scale	new population factor	whole energy above that scale
$J \asymp N^{1/4}$	$N^{-3/4}$	$N^{3/4}$
$J \asymp N^{1/2}$	$N^{-1/2}$	$N^{1/2}$
$J \asymp N^{3/4}$	$N^{-1/4}$	$N^{1/4}$

Each entry also has the displayed exponential saving in $\sqrt{(1-\alpha)\log N}$. These are intermediate range estimates, not an arbitrarily small global growth exponent.

Which part uses the actual Möbius source? For coefficients $|a_n| \leq 1$, positivity of the entries of the Green matrix gives the generic prefix bound $R_a(L) \leq 2L$. The same cutoff argument then gives a geometric envelope $S_{a,N}(J) \ll J/N$. The extra exponential saving in (4) uses the actual arithmetic bound (1). Its content is the cancellation already known in the completed Möbius prefixes, transported into a stronger mode estimate. No new multiplicative cancellation theorem is claimed for that input.

The coherent control $a_n = 1$ makes the distinction explicit:

$$R_+(L) = \sum_{a,b \leq L} \frac{1}{\max(a,b)} = 2L - \sum_{n \leq L} \frac{1}{n}.$$

It cannot satisfy the arithmetic exponential saving for any fixed positive b . Moreover, with $I = \int_0^1 \sin(\pi x/2) x^{-1} dx > 0$,

$$S_{+,N}(1) \sim \frac{2I^2}{N}, \quad \lambda_{1,N} S_{+,N}(1) \sim \frac{8I^2}{\pi^2} N.$$

These follow by a Riemann sum, using the continuous value $\pi/2$ of the integrand at zero. Its diagonal is $O(N^{-2})$, so its off-diagonal is positive of order N^{-1} . Thus the proposed closing bound requires the arithmetic source; magnitudes and the horizon geometry alone do not supply it.

5 What the estimate pays and where this mechanism stops

For any $0 < \eta < 1$, Theorem 3.1 gives

$$J \geq \lceil N^{1-\eta} \rceil \implies \lambda_{J,N} S_N(J) \leq CN^\eta, \quad E_N^{\geq J} \leq CN^\eta.$$

This pays the *entire* upper mode tail with one allowance, without an additional number-of-bands factor. Generic Parseval alone required a cutoff of order $N^{1-\eta/2}$ for the same power tolerance; the new range extends to order $N^{1-\eta}$.

The exact remaining loss in the mode estimate is visible by comparing it to the required tolerance:

$$\frac{(J/N)e^{-c\sqrt{\log(eN/J)}}}{J^2 N^{-2+\eta}} = \frac{N/J}{N^\eta} e^{-c\sqrt{\log(eN/J)}}.$$

At $J \asymp N^\alpha$, the remaining factor is $N^{1-\alpha-\eta+o(1)}$. Thus a fixed positive mode exponent α does not by itself settle every requested positive η .

For the very smallest cutoffs, the balancing source length $L_* = (N/J)e^{c\sqrt{\log(eN/J)}}$ can exceed N . In that case the proof uses $L = N$ and recovers the existing bound on $R(N)$. That is the exact step where this method supplies no smaller-horizon gain.

The mixed term remains explicitly

$$\Gamma_{N,J}(L) = \text{Re} \left\langle \Pi_J V_N^* c^{\leq L}, \Pi_J V_N^* c^{> L} \right\rangle.$$

It was bounded above by the product of the two norms. No negative sign or fixed cancellation percentage was substituted. Improving that treatment, or obtaining stronger estimates on the lowest modes of the actual prefix, is the specific unresolved arithmetic step in this mechanism. The already established counterexamples for partial prime palettes and individual prime updates remain fully compatible with this calculation.

The selected route estimates (PE) directly. The earlier continuous-field replacement, the inverse constant below 68, and the projected kernel remain available; none is charged as a new condition after this source estimate. No extra factor from $\sum |\Lambda(d) - 1|/d^{(1+\eta)/2}$ is present in the proof, because it truncates the completed source in the integer index rather than applying an absolute-value induction over dilation terms.

6 Remaining range of (PE) and complete return

Let $\ell_N = \log(eN)$, $J_0(N) = \lceil N/\ell_N^2 \rceil$, and for $0 < \eta < 1$ define

$$J_\eta(N) = \min\{J_0(N), \lceil N^{1-\eta} \rceil\}.$$

It now suffices to prove the same source comparison (PE) on the following range: for every $0 < \eta < 1$, there exists $C_\eta < \infty$ such that

$$\boxed{(\mathcal{O}_{N,J})_+ \leq C_\eta J^2 N^{-2+\eta} \quad (N \geq 2, 1 \leq J < J_\eta(N)).} \quad (5)$$

The constants must be independent of both N and J , and the coefficients are the actual $\mu(n)/n$. This is a restriction of the existing target to its still-uncontrolled range. All larger cutoffs are paid by the proved estimates, with the same permitted subpower tolerance.

Here is the full return, including all finite horizons. Write $A_\eta = \pi^2/2 + C_\eta$. On $J < J_\eta$ the diagonal allowance gives

$$S_N(J) \leq A_\eta J^2 N^{-2+\eta}, \quad \lambda_{J,N} S_N(J) \leq \frac{9}{4} A_\eta N^\eta.$$

Abel summation on these low modes therefore costs at most $\frac{9}{4} A_\eta [1 + 2 \log(3N)] N^\eta$. When this range is empty its contribution is zero. For the remaining entire energy tail, Theorem 3.1 gives

$$E_N^{\geq J_\eta} \leq C \frac{N}{J_\eta} \leq C \max\{N^\eta, \ell_N^2\} \leq C(N^\eta + \ell_N^2).$$

The two cases correspond to which term defines J_η ; no passage between selected horizons or exceptional sets is needed. Thus

$$\boxed{R(N) \leq \frac{9}{4} A_\eta [1 + 2 \log(3N)] N^\eta + C(N^\eta + \log^2(eN)).} \quad (6)$$

For a desired $0 < \delta < 2$, use $\eta = \delta/2$ and absorb logarithms into $N^{\delta/2}$. Larger exponents then follow as well. This yields subpower $R(N)$, and the original terminal gives

$$|M(N)| \leq \sqrt{NR(N)} = O_\delta(N^{1/2+\delta/2}).$$

The classical Möbius criterion then supplies RH. The base $N = 1$ is $R(1) = 1$; real cutoffs use $M(x) = M(\lfloor x \rfloor)$.

Current status. Theorem 3.1 and its range improvement are proved. The comparison (5) is not proved here. There is no further energy, tail, or regularity condition after that comparison.

7 Compact diagnostics and exact scope of the checks

The computation tests Theorem 2.1 at $N = 64, 257, 1024, 4096$ and $J = 1, 3, 8, 32$, taking $L = \min(N, \lceil 2N/J \rceil)$ for a reproducible finite split. This chosen test cutoff is not a numerical fit of A, b in the asymptotic theorem. All quantities use the exact finite sine transform and the exact historical-plus-terminal prefix energy.

N	J	L	$\lambda_J S_N(J)$	$E_N^{\geq J}$	common allowance
64	8	16	0.169033	0.749076	2.343828
257	8	65	0.093853	1.193002	4.288237
1024	32	64	0.090570	1.183036	4.595958
4096	32	256	0.035052	1.396416	9.851200

The last column is exactly $(\sqrt{R(L)} + \sqrt{\lambda_J T_{L,N}})^2$, evaluated numerically. The finite allowances have slack. Their role is to test the inequality and indexing; the growing-range improvement is the analytical optimization in Section 3 with the independently established arithmetic bound.

The projected prefix–tail cross term has both signs in these tests. For example $\lambda_J \Gamma_{N,J}(L)$ is approximately -0.0154691 at $(N, J, L) = (64, 8, 16)$ and $+0.0209030$ at $(64, 32, 4)$. These are finite observations and were not used as a universal sign input. The full signed identity $S = \|\Pi_J h\|^2 + \|\Pi_J t\|^2 + 2\Gamma$ is checked separately.

The all-positive control at $N = 257, J = 1$ gives

$$S_+(1) = 0.01452923, \quad \mathcal{D}_+(1) = 0.00005744, \quad \mathcal{O}_+(1) = 0.01447179,$$

and $R_+(257) = 507.871764$. It satisfies the general truncation inequality with its own prefix energy, and its asymptotic first-mode energy coefficient is $8I^2/\pi^2 \approx 1.52305105$. The analytic calculation in Section 4 establishes the coherent obstruction without numerical extrapolation.

The source package includes `checks.py` and complete `checks.json`. The code uses NumPy and SciPy; it is floating-point diagnostic code, not an interval certificate. No broad rerun to 10^7 was performed. Claude’s reported tests through sampled horizons up to 10^7 remain reported data; their underlying scripts, horizon list and normalization were not inspected. Ng’s conditional benchmark is not used as an independent arithmetic input.

References and resume record

Supplied checkpoint. *Arithmophysics: Shared Checkpoint and Continuous-Field Return*, six-page user-supplied continuation. It fixes (PE), the source diagonal, the original-metric energy, the independent baseline (BL), and the completed status of the earlier diagnostics.

Prior project papers.

- *Arithmophysics: Prime Modes and Complete Packet Cancellation*, 13 pages. The sine spectrum, uniform gain bounds, Abel criterion, terminal energy, and unconditional baseline used here.
- *Arithmophysics: Complete Fluctuation Packets*, 11 pages. The constant density-discretization allowance, the inverse below 68, and the complete signed fluctuation checkpoint, retained as established tools.

External inputs.

- NIST DLMF, equation 27.11.12, <https://dlmf.nist.gov/27.11.E12>. The unconditional Möbius-sum estimate gives the existing energy bound (1). This source is not used for a new multiplicative correlation theorem.
- Lynnelle Ye, *Bounding sums of the Möbius function over arithmetic progressions*, 2014, <https://arxiv.org/abs/1406.7326>. The introduction states the classical equivalence used after obtaining the full Möbius growth bound in the conditional closing deduction.

The source truncation inequality and its optimized consequence are derived in full here. No historical novelty claim is made for the underlying Hilbert-space inequalities, Fourier basis, or unconditional number-theoretic input.

Resume from this precise result. The independently proved bound is

$$\max\{\lambda_{J,N}S_N(J), E_N^{\geq J}\} \leq \min\{ANe^{-b\sqrt{\log(eN)}}, C(N/J)e^{-(b/2)\sqrt{\log(eN/J)}}\}.$$

It provides a quantified improvement for growing modes, with every horizon covered. The source prefix is a completed integer prefix, and its terminal is retained. The coefficient tail costs $\lambda_J T_{L,N}$ in the relevant weighted comparison. The mixed term is bounded, never deleted.

For each requested exponent η , the remaining comparison is exactly (PE) on $J < \min(J_0, \lceil N^{1-\eta} \rceil)$. Section 6 gives the full return if that comparison is established. The current work does not prove a new global growth exponent for $R(N)$ and does not prove the subpower target.

The source ZIP contains the manuscript and PDF, the finite checks and their output, a concise checkpoint, and all three supplied/prior input PDFs.